Technical pack
The Operator 35 L

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A matte black 72-hour emergency backpack for adults, sold worldwide. The factory makes the empty bag with its internal pouch set, sleeves and labels; contents are packed by us in the UK. **The brief: the lowest cost that is still strong, durable and waterproof.** Use the factory's standard value materials and parts wherever they meet our strength tests (Sheet 5). Keep the look clean: everything black, neat edges, a structured shape (page 2). Please also suggest any further savings that keep the bag strong.

SHEET	CONTENTS	WHAT IT COVERS
1	**Exterior views**	520 H × 320 W × 220 D mm, 35 L. Name window by the handle, left bottle pocket, Morse SOS on the right panel, hip belt, whistle sternum buckle, grab handle, compression straps, reflective strips
2	**Interior and pouch set**	Lid mesh pocket, elastic loop row, card sleeve, A5 document sleeve, four colour-coded pouches with factory-printed names, sewn-through pouch rails with press-stud straps
3	**Docking system**	Four female buckles on the front, beside the pocket and anchored to internal load straps, so our future child, baby and pet bags clip on. Must be built into every Operator now
4	**Branding, hardware, materials**	Rubber patch 70 × 40 mm, Morse print, blue tag, name card, inside label, colours, full material specification
5	**Bill of materials, QC, packing**	Quantities per bag, construction standards, pull, drop and water tests, packing, terms

What we need from you

1. **Unit price for 300 Operators**, FOB and DDP London, in the value specification on Sheet 4.
2. Price per bag for each option, quoted separately: (a) YKK AquaGuard on the main zip; (b) laser-cut Morse panel instead of print; (c) moulded rubber zip pulls with our logo; (d) full reflective piping; (e) 900D shell instead of 600D; (f) a 50 mm padded hip belt.
3. One pre-production sample: cost, lead time, and whether the fee is refunded against the order.
4. Production lead time for 300 pieces after sample approval.
5. One-off costs: rubber patch mould and screen for the Morse print.
6. Swatches of your standard 600D black polyester (2 or 3 options), with REACH compliance.

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Value build, clean finish: the checklist

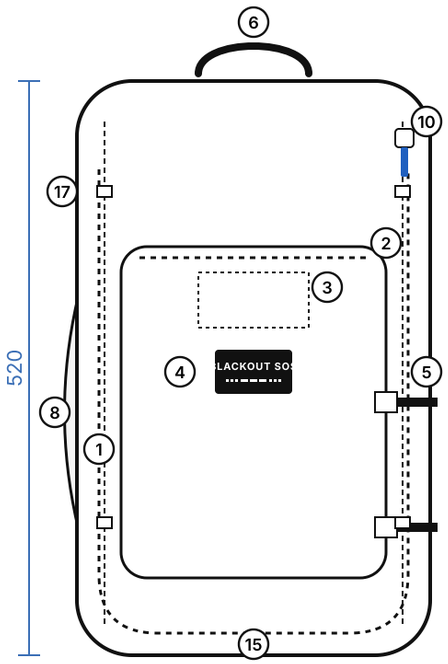
Every point below is a pass/fail item at sample approval and at pre-shipment inspection.

1. **One true black.** Shell, base, webbing, thread, zip tape, sliders, buckles and lining all match under daylight: no brown, blue or grey cast between parts. The only colour on the outside is the blue tag and side loop.
2. **Matte and waterproof.** 600D polyester with the waterproof TPU layer on the inside, so the outside face stays matte; seams taped.
3. **Holds its shape.** 10 mm EVA back panel and a 2 mm PE stiffener in the front panel so the bag stands upright on its own and the front stays flat with clean lines, empty or full.
4. **Brand details.** The black rubber front patch and the gloss-black Morse SOS print on the matte side panel.
5. **Black hardware.** Black zip sliders and black acetal buckles; nothing silver.
6. **Clean edges.** All internal seams bound with 20 mm tape, webbing ends heat-cut and folded under, no loose threads, no exposed foam, straight even stitching at 8–9 stitches per inch, double-needle on straps.
7. **Reflective strips** on both shoulder straps, so the wearer shows up in a torch beam.
8. **Low-profile docking buckles** on short black tabs beside the front pocket, neat and symmetrical, so the front still reads as one clean panel.

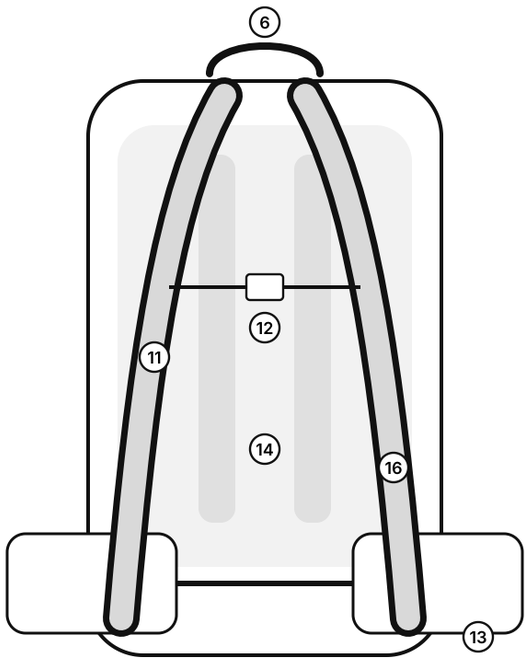
Concept render for look, finish and proportion only. All dimensions, materials and construction are as drawn on the sheets that follow. No camouflage, no visible MOLLE, no bright branding.

Sheet 1 — Operator 35 L · exterior views

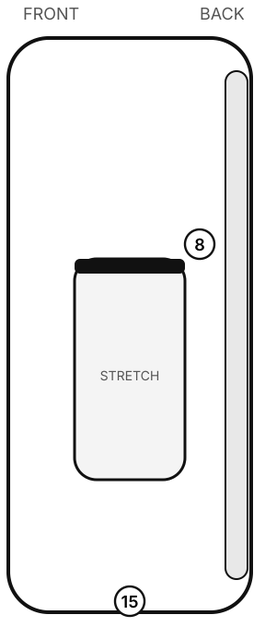
Scale 0.6 px per mm on screen · all dimensions in mm · colour: black throughout, tag blue



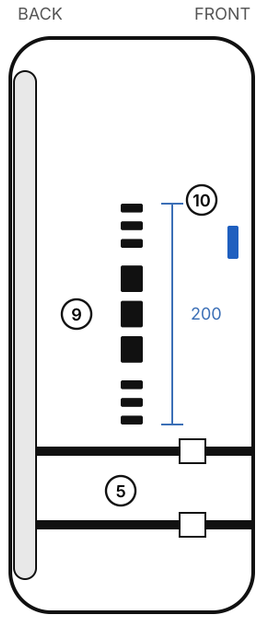
FRONT



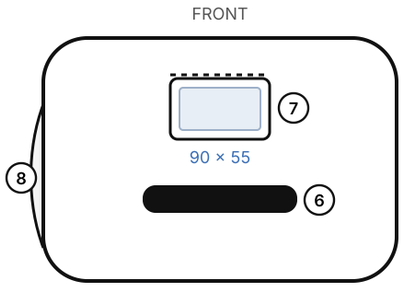
BACK



LEFT SIDE



RIGHT SIDE



TOP

Overall
H 520 × W 320 × D 220 mm · 35 L nominal, 31 L usable
Empty weight target 1.3 kg · load rating 10 kg
Carry-on size: fits 55 × 40 × 23 cm

Build
Shell 600D polyester, matte, TPU laminated, waterproof, black
Base 600D double layer · lining 210D polyester
Main zip #8 waterproof coated coil, 2 sliders, storm flap
Pocket zip #5 coated coil · black acetal buckles
Seams: taped on main body · bar-tacks at every strap anchor

1 Clamshell main opening, #8 waterproof coated (PU) coil zip, two black sliders, 20 mm storm flap over the zip

2 Front organiser pocket 240 × 300 mm, #5 coated coil zip, mesh inner

3 Loop-Velcro field 100 × 50 mm, black, for the customer's own patch

4 BLACKOUT SOS rubber patch 70 × 40 mm, black-on-black, sewn (Sheet 4)

5 Compression straps, right side only: 20 mm webbing, 20 mm side-release buckles, at 290 and 400 mm from top

6 Grab handle: 25 mm webbing, 140 mm, padded sleeve, bar-tacked both ends

7 Name-tag window 90 × 55 mm on top panel beside the handle: clear TPU window pocket with a slot opening and flap (no zip), holds an 85 × 54 mm card

8 Side bottle pocket, left: 100 × 200 mm heavy stretch mesh, elastic cuff, drain eyelet, fits 750 ml to 1 L

9 Morse SOS on the right side panel, 200 mm tall: gloss black screen print on the matte shell, black on black (Sheet 4). Laser-cut slots with mesh backing quoted as an option

10 Colour tag 60 × 15 mm looped on the main zip pull, plus 10 × 30 mm loop on the right side; blue (Sheet 4)

11 Shoulder straps 60 mm, S-curve, 10 mm EVA foam, mesh face, 40 mm ladder-lock adjusters

12 Sternum strap 20 mm on rail, buckle with built-in whistle

13 Hip belt: 40 mm webbing, padded wings 150 × 90 mm, stows behind the back panel

14 Back panel: 10 mm EVA, two airflow channels, mesh face

15 Base panel 600D polyester, double layer, PU coated, no feet

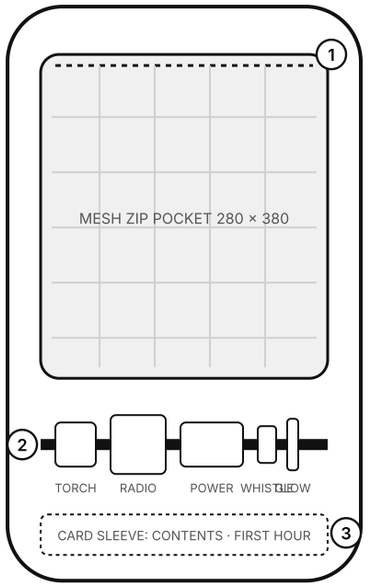
16 Black reflective strip 10 × 60 mm on each shoulder strap (full reflective piping quoted as an option)

17 Docking points (dashed): four 20 mm female side-release buckles on short 25 mm webbing tabs, **beside** the front pocket, never on it: centres 270 mm across and 300 mm down. Each pair is sewn through the front panel into a vertical 25 mm load strap inside the panel that runs from the top seam to the base seam and is caught in both, so the weight goes into the bag body like a shoulder strap. Takes one of our future child, baby or pet bags at a time, 10 kg max (Sheet 3)

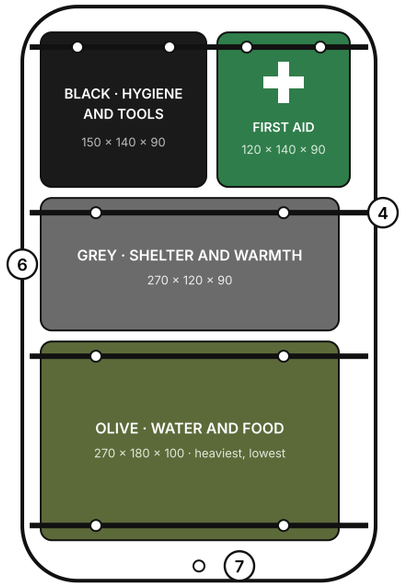
Not shown: hydration port omitted; no external MOLLE; all hardware black; 2 mm PE stiffener inside the front panel; all thread black.

Sheet 2 — Operator interior, packing map and pouch set

Lining black 210D polyester ripstop



LID (inside of front panel)



MAIN BODY (back wall)

4 rails · O press studs · A5 document sleeve sewn flat behind the pouches

Interior callouts

1 Lid mesh pocket 280 × 380 mm, black polyester mesh, #5 coil zip across the top, holds the lid items and the cards

2 Elastic loop row: 25 mm black elastic, five loops sized torch · radio · power bank · whistle · glow sticks, bar-tacked between loops

3 Card sleeve 150 × 60 mm, clear TPU, under the loop row: contents card and first-hour guide

4 Pouch rails: four rows of 25 mm nylon webbing across the back wall, one at the top edge of each pouch row plus one at the bottom of the olive pouch, bar-tacked every 50 mm to make loops. Each rail is sewn through the shell back panel with a 40 × 20 mm reinforcing patch behind every bar-tack, and its ends are caught in the side seams; never sewn to the lining alone. Each pouch has 25 mm webbing straps on its back that thread behind a rail loop and close with a black metal press stud (15 mm, ring-spring): 2 straps on black, green and grey; 4 on olive (2 top, 2 bottom) because it is the heaviest. Every pouch clips in and out on its own. No hook-and-loop

5 Document sleeve A5 (148 × 210 mm), TPU-lined, Velcro flap, sewn flat to the back wall behind the pouches

6 Packing map, top to bottom: black (hygiene and tools) beside green (first aid) at the top for quick access; grey (shelter and warmth) in the middle; olive (water and food, about 3 kg) at the bottom, against the back panel. Nothing loose: all soft items go in the grey pouch

7 Drain eyelet, 8 mm brass, black, in the base

8 Main compartment lining 210D polyester ripstop, black; lid lining the same with the mesh pocket over it

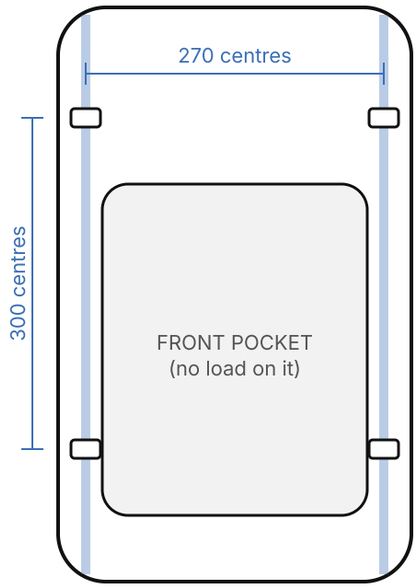
9 Two 20 mm webbing loops on the back wall at the top for the hydration bladder hook (no port, loops only)

Pouch	Colour	Size mm (W × H × D)	Holds	Construction
Water and food	Olive 210D ripstop	270 × 180 × 100	3 × 500 ml water pouches, 2 × 500 g ration blocks, comfort pack, tablets, spork (about 3 kg)	Gusseted pouch (depth as listed), #5 coil zip across the top only, 20 mm grab tab, pouch name screen-printed by the factory on the front in white, 8 mm capitals (WATER AND FOOD · SHELTER AND WARMTH · HYGIENE AND TOOLS · FIRST AID), plus a 60 × 20 mm clear label window for our contents card, 25 mm press-stud straps on the back for the rails: 2 per pouch, 4 on the olive pouch (item 4), lining 210D black
Shelter and warmth	Grey 210D ripstop	270 × 120 × 90	Bivvy, poncho, 2 foil blankets, 4 hand warmers, hat, gloves, socks	
Hygiene and tools	Black 210D ripstop	150 × 140 × 90	Wipes, sanitiser, toilet roll, toothbrush, pads, bags, multi-tool, tape, cord, ties, ferro rod, pen	
First aid	Green 210D with white cross	120 × 140 × 90	First aid kit contents, trauma dressing, medication pouch, masks, gloves	
Lid mesh pocket	Black mesh	280 × 380 flat	Head torch, radio, power bank, cable, whistle, glow sticks, batteries	Sewn into the lid lining, #5 coil zip
Document sleeve	Black TPU-lined	A5, 160 × 220	Document pouch, map, contacts card, cash	Sewn to the back wall, Velcro flap

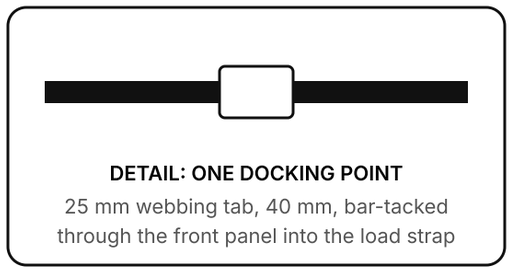
Sheet 3 — Docking system (piggyback points)

Scale 0.6 px per mm · dimensions in mm

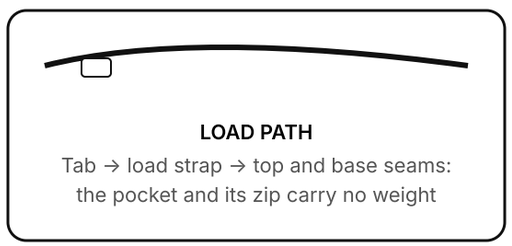
Every Operator carries four female buckles on its front, beside the front pocket, never on it. Our smaller bags, coming later (child, baby and pet), have four matching male buckles on their backs and clip straight on, so one adult carries two bags hands-free. **These buckles must be built into all 300 Operators now**; the smaller bags will be made to this exact spacing.



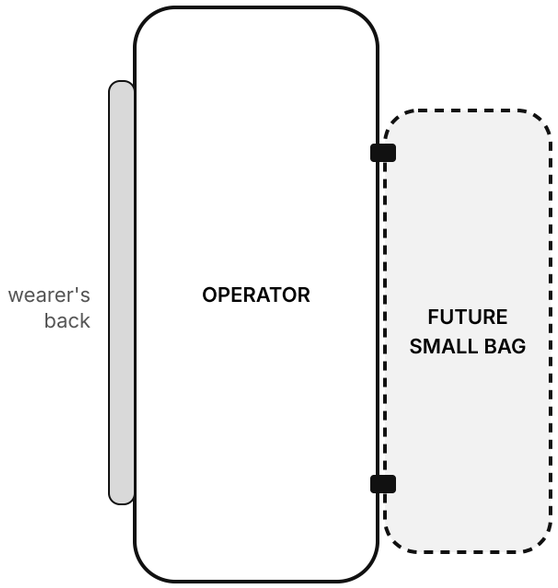
OPERATOR · FRONT
4 × 20 mm FEMALE buckles beside the pocket
Blue: internal load straps, top seam to base seam



DETAIL: ONE DOCKING POINT
25 mm webbing tab, 40 mm, bar-tacked through the front panel into the load strap



LOAD PATH
Tab → load strap → top and base seams: the pocket and its zip carry no weight



SIDE VIEW, DOCKED
Small bag clips to the Operator's outer face
(dashed: not part of this order)

Rules

One docked bag at a time
Max docked load 10 kg
Spacing ±2 mm on every bag: future bags must fit all Operators made now

Tests on the Operator

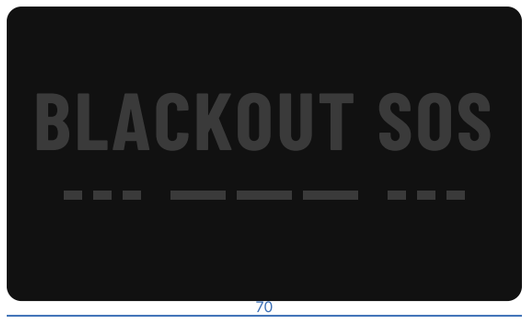
Each tab: 25 kg pull; all four together 40 kg, no stitch failure
Buckle: 500 clip/unclip cycles
Every bag checked on a 4-buckle jig (below)

Please make a simple test jig (a flat board with four 20 mm male buckles at 270 × 300 mm centres) and use it to check docking on every Operator at inspection; send one jig with the sample.

Sheet 4 — Branding, hardware and materials

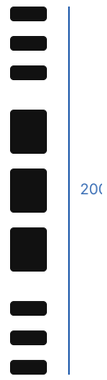
Artwork drawn at 4 px per mm · wordmark font Barlow Condensed Bold, all capitals · factory to supply strike-offs of every item

A · Front patch 70 × 40 mm



Matte black PVC rubber patch, 2 mm base, letters and Morse raised 1 mm in the same black; the mark reads only by its sheen. Cap height 9 mm. Morse: dots 2.5 × 1.2 mm, dashes 7.5 × 1.2 mm, 1.5 mm between marks, 4 mm between groups, 4 mm below the letters. Corner radius 2 mm. Sewn on with black thread 2 mm from the edge.

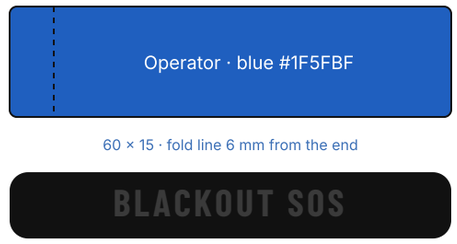
B · Morse SOS print, side panel



Bag	Slot W	Dot	Dash	Gap / group	Total
Operator	20	8	24	8 / 16	200

All in mm. Base: screen-printed in gloss black ink on the matte shell, black on black, edges sharp. Option: laser-cut slots with black mesh backing. Centred on the panel width; top of pattern 150 mm from the top seam.

C · Colour tag and zip pull



Tag: woven polyester, high density, 60 × 15 mm, colour both faces, folded and lark's-headed through the slider so the customer can swap it. Side loop: 10 mm webbing in the same colour, 30 mm loop, bar-tacked. Zip pulls: black cord with black end cap on every slider (moulded rubber pull with wordmark, as drawn, quoted as an option).

D · Name card 85 × 54 mm (printed, supplied by us)

350 gsm card, matt laminated, write-on with a ballpoint. Ships in the window; two spares in the lid sleeve.

E · Inside label 60 × 40 mm (woven, sewn into the back wall seam)

Black ground, white and grey weave. Label reads OPERATOR 35 L · MAX LOAD 10 kg. No QR code. A separate care and composition label to UK textile rules sits beside it.

F · Colours

Shell, webbing, hardware, thread — black. Match the shell swatch we approve; all black parts the same black; no blue or brown cast.

Blue tag, Operator — #1F5FBF, close to Pantone 2935 C

Olive pouch — #5C6A3A · **grey pouch** #6B6B6B · **green first-aid pouch** #2F7D4A with white cross

Pantone references are the nearest match for the factory's lab-dip; we approve on physical swatches, not screen. Tags must pass 20 washes without bleeding onto the black shell.

Material or hardware	Specification	Where	Accepted alternative
Shell fabric	600D polyester, black, matte face, TPU laminated on the back, waterproof: water column 3,000 mm or better	All outer panels	Your standard 600D black; send swatches
Base fabric	600D polyester, double layer, PU coated	Base panel	900D single layer
Lining	210D polyester, PU coated, black; pouches dyed olive, grey, green, black	Compartments, pouches	190T polyester
Mesh	Polyester air mesh (back panel, strap faces); flat polyester mesh (lid pocket, bottle pocket)	Back, straps, lid	Your standard
Foam	EVA or PE closed-cell, 8–10 mm back and straps, 5 mm hip belt	Padding	Your standard
Stiffener	PE sheet 2 mm in a sleeve in the front panel	Front panel	None: keeps the shape
Zips	Waterproof PU-coated coil on all external zips (#8 main with storm flap, #5 front pocket); plain #5 coil inside; black sliders	All zips	None needed
Zip pulls	Black cord with black end cap, stock part	Every zip	Option: moulded rubber pulls with logo
Buckles and adjusters	Black acetal, your standard grade: 20 mm side-release (whistle type on sternum; 4 female docking), 25 mm ladder locks, 40 mm hip belt	As Sheet 1	Must pass the Sheet 5 load tests
Webbing	Nylon or polyester, black, tight weave, 20, 25 and 40 mm; ends heat-cut and folded	All	Polypropylene not accepted on load-bearing straps
Reflective	Two reflective strips 10 × 60 mm, one on each shoulder strap	Straps	Option: full piping
Windows	Clear TPU or PVC 0.5 mm	Name window, card sleeve, pouch label windows	None
Thread	Bonded polyester #69, black; heavier on handle	All	None
Seams	Seam tape on all shell-to-shell seams of the main compartment; storm flap over the main zip; internal edges bound	Main body	Welded seams if you offer them

Sheet 5 — Bill of materials, construction, QC and packing

Quantities per bag · factory to confirm consumption from its own patterns

COMPONENT	PER OPERATOR
Shell: 600D polyester, matte, TPU laminated, black, water column 3,000 mm+	1.2 m²
Base: 600D polyester, PU coated, double layer	0.1 m²
Lining: 210D polyester ripstop, black · pouch fabric dyed olive, grey, black, green	1.4 m² + 0.6 m²
Stretch mesh (bottle pocket)	1 pocket
3D spacer mesh (back, straps) + flat mesh (lid pocket, Morse backing)	0.25 + 0.2 m²
EVA foam 10 mm (back panel, 2 straps) · 5 mm (hip belt, grip)	as listed
PE stiffener 2 mm, front panel, sewn into a sleeve	1
Main zip #8 waterproof coated coil, 2 black sliders, storm flap	1,100 mm
Pocket zip #5 waterproof coated coil (front pocket)	2
Internal zips #5 coil (lid mesh pocket; 4 pouches, top only)	1 + 4
Zip pulls: black cord with black end cap (stock part)	8 (4 external, 4 pouches)
20 mm side-release buckles, matte black acetal	2 compression + 4 female docking
20 mm sternum buckle with built-in whistle	1
25 mm ladder locks · 40 mm hip-belt buckle	2 · 1
Webbing, black nylon: 20 mm · 25 mm · 40 mm	1.5 m · 2.3 m (incl. 4 pouch rails, docking tabs, 2 internal load straps) · 1.2 m
Elastic 25 mm · reflective strips 10 × 60 mm	0.4 m · 2
Press studs 15 mm, matte black metal, ring-spring (pouch straps)	10 (olive 4, others 2 each)
Clear TPU 0.5 mm: name window 90 × 55, lid card sleeve, pouch label windows	1 + 1 + 4
Pouches, names screen-printed in white: olive 270 × 180 × 100, grey 270 × 120 × 90, black 150 × 140 × 90, green with white cross 120 × 140 × 90	4
Document sleeve A5, TPU-lined, Velcro flap	1
Rubber patch 70 × 40 · loop field 100 × 50 · blue tag 60 × 15 · blue side loop	1 each
Labels: inside woven · care and composition · drain eyelet 8 mm	1 · 1 · 1
Name card 85 × 54 mm	supplied by us

Construction

Lockstitch 8–9 stitches per inch; double-needle on straps; #138 thread on the handle. 42-stitch bar-tacks at strap tops and bottoms, handle ends, hip-belt anchors, compression straps, pouch-rail loops and docking tabs, each through the shell with a reinforcing patch. Docking tabs are sewn through the front panel into two vertical 25 mm load straps inside the panel, caught in the top and base seams; nothing loads the front pocket. Main-compartment shell seams taped on the inside before lining; all internal raw edges bound with 20 mm tape. Storm flap 20 mm over every external zip. Morse SOS screen-printed in gloss black, 200 mm tall, centred on the right panel. Webbing ends heat-cut and folded under.

Tests (sample and pre-shipment)

Water (the bag is sold as waterproof): closed bag under a shower for 10 minutes, then left standing in 20 mm of water for 5 minutes; no water inside the main compartment. Load: handle 30 kg static for 5 minutes; each strap anchor and docking tab 25 kg pull; all four docking tabs together 40 kg for 5 minutes with the front pocket and zip undamaged; each pouch strap and rail loop 15 kg for 1 minute, no stud release. Drop: packed to 7 kg (olive pouch 3 kg), 5 drops from 1 m on each face, every pouch still attached. Zips: 2,000 cycles on the main zip. Press studs 1,000 cycles; docking buckles 500 cycles. Whistle audible at 1 m (100 dB). Tag colour-fast after 20 washes. Dimensions ±5 mm; empty weight 1.3 kg ±5%. Colour: all black parts match in daylight. Pre-shipment inspection by a third party to AQL 2.5 major / 4.0 minor.

Packing

Each bag: straps tucked, name card in its window, two spare cards in the lid sleeve, pouches flat inside and unfilled, in a recycled poly bag with a silica-gel sachet; no paper stuffing. 10 bags per 5-ply export carton (30 cartons for 300), carton under 20 kg, labelled BLACKOUT SOS · OPERATOR · BLUE · quantity · carton x of 30 · weight · dimensions · barcode.

Terms

Order: 300 Operators. Quote FOB Xiamen or Shenzhen and DDP London; HS 4202.92. Samples: swatch card and hardware set → pre-production sample → gold-seal sample signed by us and kept by the factory. Payment: 30% deposit, 70% after passed inspection. We supply vector artwork for the patch, tag, zip pull, labels and name card with the order to sample.